

A Graph Foundation Model for Unified Anomaly Detection

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Abstract

Graph anomaly detection (GAD) identifies nodes, edges, and sub-graphs that deviate from normal patterns, playing a key role in cybersecurity, finance, and web technologies. However, existing deep learning paradigms trained on fixed datasets struggle to generalize to unseen graphs, while current graph foundation models (GFMs) face inherent challenges in anomaly detection. Their generalization objectives conflict with the rarity and divergence of anomalies, and existing attempts to develop GFMs for GAD mainly fall into two directions: single-task cross-domain models and domain-specific multi-task models. The former transfers knowledge across domains but fails to capture cross-level dependencies, whereas the latter handles multiple tasks jointly but lacks cross-domain adaptability. Both rely heavily on labeled data, limiting scalability. These limitations highlight the need for a unified unsupervised graph foundation model capable of generalizing across domains and detecting anomalies at multiple levels. We propose GFM-UAD, a Graph Foundation Model for Unified Anomaly Detection. It is pretrained on a single dataset and performs multi-level anomaly detection across unlabeled domains. To address anomaly scarcity, GFM-UAD extracts high-confidence normal samples and synthesizes diverse anomalies through adversarial generation. By freezing the backbone and fine-tuning the detection head with distribution alignment and message passing, the model adapts flexibly to unseen environments without supervision. Experiments across multiple datasets demonstrate that GFM-UAD achieves superior accuracy and generalization, establishing a foundation for future unsupervised graph foundation models in anomaly detection.

CCS Concepts

• **Computing methodologies** → **Neural networks**; *Anomaly detection*.

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Keywords

Graph Neural Network; Unsupervised Learning; Graph Anomaly Detection; Graph Foundation Model

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1 Introduction

Graph Anomaly Detection (GAD) [10, 35, 57, 58] aims to identify samples such as nodes, edges, or entire graphs [24–26, 32] that deviate from the universal patterns of normality. This capability is critical in many domains. In cybersecurity, GAD is employed to detect abnormal network traffic [12] and interactions [9, 46, 48]. In financial systems, it helps identify fraudulent transactions [15, 20, 43]. In web data analysis, it facilitates the discovery of irregular patterns in data structures. As real-world graph data grows exponentially in both volume and complexity, traditional deep learning paradigms [3, 16] that depend on task-specific datasets are becoming increasingly insufficient. They struggle to handle previously unseen graphs and therefore fail to generalize effectively to practical and dynamic environments.

In recent years, the success of Graph Foundation Models (GFMs) [31, 51, 60] has inspired the pursuit of more general and adaptable graph learning frameworks. Instead of relying on single datasets or narrowly defined tasks, GFMs aim to learn transferable representations through pretraining across diverse graph structures and tasks, improving their adaptability to new domains. However, integrating GFMs with Graph Anomaly Detection remains highly challenging. Anomaly detection focuses on rare and distributionally divergent patterns, which fundamentally conflict with the generalization objectives of GFMs. This discrepancy makes it difficult to build a unified model that can effectively detect anomalies in a consistent and robust way.

Existing efforts to design GFMs for GAD can be broadly classified into two categories: single-task cross-domain models and domain-specific multi-task models. The first category, represented

by UNPrompt [39], ARC [34], and AnomalyGFM [40], focuses on transferring learned graph representations across domains but remains limited to a single detection objective. In real-world scenarios, anomalies at different levels often interact. For example, an anomalous edge may arise from abnormal node behavior, while graph-level irregularities may result from the accumulation of local anomalies. Neglecting such cross-level dependencies prevents existing methods from capturing intrinsic and comprehensive anomaly structures, reducing their robustness in complex graphs. The second category, including BOURNE [30], UniGAD [28], and Uniform [45], can perform multi-level anomaly detection but lacks cross-domain adaptability. These models are typically tailored to specific datasets and thus cannot be easily generalized to unseen domains. Furthermore, both categories commonly rely on labeled data, which is often scarce and expensive due to the difficulty of annotating anomalies. In contrast, the essence of modern foundation models lies in their ability to acquire transferable knowledge through unsupervised or self-supervised pretraining. Such models learn universal representations by capturing structural regularities and contextual dependencies directly from raw data, enabling them to generalize across diverse domains without supervision. This paradigm, which has driven the success of large language models and emerging graph foundation models, highlights the potential and importance of unsupervised learning for designing GFMs for GAD.

These limitations reveal an urgent need for a unified unsupervised GFM that can perform multi-level anomaly detection while generalizing to unseen domains without relying on labeled supervision. However, two fundamental challenges arise. First, the model is pretrained on a source domain but expected to generalize to unknown target domains where anomalous samples are inherently sparse. Second, it must capture dependencies across multiple levels of tasks while being fine-tuned without labeled data.

To address these challenges, we propose a Graph Foundation Model for Unified Anomaly Detection, GFM-UAD, which is pretrained on a single dataset and can detect anomalies at multiple levels across unlabeled domains. To mitigate the scarcity of anomalies, we extract high-confidence normal samples and generate diverse synthetic anomalies through adversarial generation. This strategy enriches the diversity of anomaly patterns and enhances generalization across different domains. After pretraining, the backbone parameters are frozen, and the detection head is fine-tuned through distribution alignment using message passing, enabling flexible adaptation to new anomaly detection tasks without labels.

Our contributions are summarized as follows.

- **New Task.** To the best of our knowledge, this is the first attempt to explore a graph foundation model for unified anomaly detection in an unsupervised setting.
- **New Method.** We propose GFM-UAD, an unsupervised GFM that synthesizes anomalies through adversarial generation and fine-tunes the detection head via distribution alignment to achieve cross-domain adaptability.
- **New Performance.** Extensive experiments across diverse datasets and multi-level tasks demonstrate that our method consistently outperforms existing state-of-the-art approaches.

2 Related Work

Related work is shown in **Appendix A**.

3 Methodology

In this section, we present the proposed GFM-UAD approach in detail. Unlike existing anomaly detection methods that heavily rely on labeled anomalies or struggle to generalize across domains, GFM-UAD introduces a heuristic strategy to generate sufficient anomalous samples directly from high-confidence normal instances. This design significantly enhances its domain generalization capability for inference of unseeded samples. Moreover, instead of conventional fine-tuning that requires explicit supervision, GFM-UAD performs distribution-driven fine-tuning across different domains, ensuring adaptive and robust anomaly detection in diverse environments. The framework of GFM-UAD is composed of three key components: High-Confidence Sample Extraction, Anomaly Generation by Adversarial Network, and Task-Driven Fine-Tuning via Distribution Alignment. The overall architecture is shown in Fig. 1.

3.1 Notations

We denote a collection of graphs as $\mathcal{G} = \{G_1, G_2, \dots, G_M\}$, where each graph $G = (V, E, \mathbf{X}, \mathbf{A})$ consists of a node set V , an edge set E , a node feature matrix $\mathbf{X} \in \mathbb{R}^{|V| \times d}$, and an adjacency matrix $\mathbf{A} \in \mathbb{R}^{|V| \times |V|}$. For node $v_i \in V$, its feature vector is denoted as $\mathbf{x}_i \in \mathbb{R}^d$, where d is the dimension of node attributes. Unless otherwise specified, graphs are assumed to be undirected, in which case $\mathbf{A}$ is symmetric. In contexts involving normalized message passing, we define $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ and $\tilde{\mathbf{D}} = \text{diag}(\sum_j \tilde{a}_{ij})$. A Graph Foundation Model is represented as f_θ , parameterized by θ , and pretrained over the graph dataset $\mathcal{G}_{\text{pre}}$ using a self-supervised objective $\mathcal{L}_{\text{pre}}(\theta; \mathcal{G}_{\text{pre}})$. The pretrained model is then adapted to an inference task T with dataset $\mathcal{D}_T$ after a fine-tuning mechanism $\mathcal{A}$. Inference tasks are categorized as node-level, edge-level, and graph-level.

3.2 High-confidence Samples Extraction

In this section, we first identify normal samples by reconstruction from the single dataset and then sample high-confidence ones to train the generated-adversarial network in the next step. Specifically, we utilize GCN as encoder $\mathcal{F}_e$ to project the raw attribute to the latent space for each graph, as

$$\mathbf{H} = \mathcal{F}_e(\mathbf{X}, \mathbf{A}), \quad (1)$$

where $\mathbf{H}$ is the node embeddings. To assess the normality of each graph, we employ a lightweight reconstruction head $g_a(\cdot)$ and $g_e(\cdot)$ to recover the node embeddings $\mathbf{H}$ back to its origin attributes $\mathbf{X}$ and adjacent matrix $\mathbf{A}$, respectively, as

$$\mathbf{X}' = g_a(\mathbf{H}), \quad \mathbf{A}' = g_e(\mathbf{H}\mathbf{H}^\top), \quad (2)$$

where $\mathbf{X}'$ and $\mathbf{A}'$ are the reconstructed node attribute matrix and adjacent matrix, respectively. Based on these reconstructions, the anomaly score of node i can be defined by combining the attribute reconstruction error and the structural reconstruction error:

$$s_i = \|\mathbf{x}_i - \mathbf{x}'_i\| + \|\mathbf{a}_i - \mathbf{a}'_i\|. \quad (3)$$

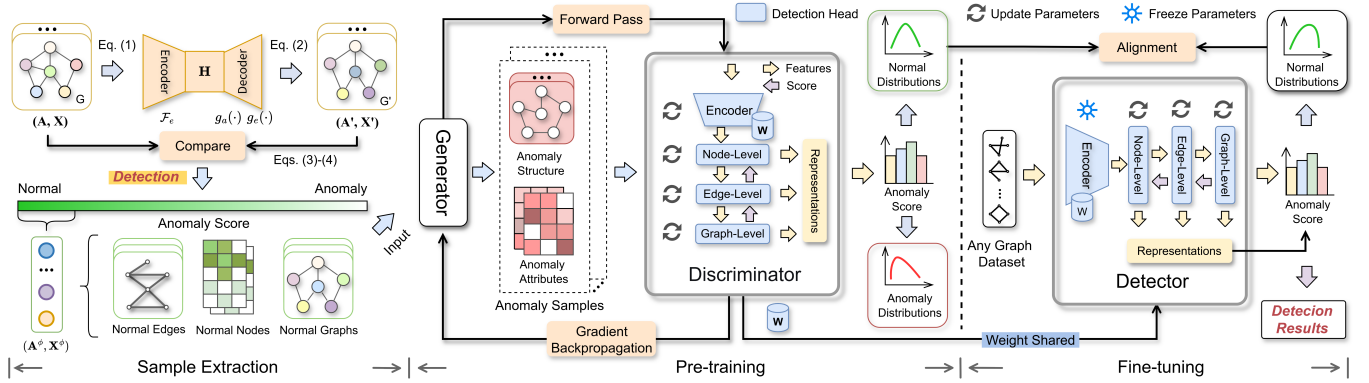


Figure 1: The overall architecture of GFM-UAD model. The left part represents the selection of high-confidence normal samples, the middle part corresponds to the pre-training process, and the right part illustrates the fine-tuning and inference process.

For finer granularity, the edge-level reconstruction score can be expressed as $s_{ij}^\phi = \|a_{ij} - a'_{ij}\|$, which captures the deviation between the original and reconstructed connections. The graph-level anomaly score is a comprehensive score of all edge-level and node-level information. Finally, to ensure robustness, we rank all nodes, graphs, and subgraphs based on their anomaly scores and select those $\theta\%$ with the lowest scores as high-confidence normal samples. The node attributes $\mathbf{X}^\phi$ and graph structures $\mathbf{A}^\phi$ of these selected instances are then used to guide the adversarial learning stage, effectively mitigating the influence of anomalous or noisy data during model optimization.

3.3 Anomaly Generation by Advertised Network

In this section, we leverage high-confidence normal samples to generate multi-level anomalous samples and simultaneously train the detector. Specifically, we design an adversarial variational learning network, which consists of a Generator $\mathcal{G}$ and a Discriminator $\mathcal{D}$, which are optimized jointly in a minimax manner. The generator perturbs the latent representation of normal graphs to produce anomalous variations at different levels (i.e., node, edge, and graph), while the discriminator distinguishes real graphs from generated anomalies, thereby refining the detector's decision boundary.

For the generator $\mathcal{G}$, the objective is to synthesize anomalous graphs that are both realistic and challenging for the discriminator, while ensuring that the generated graphs deviate sufficiently from normal samples. In detail, we extract the graph latent distribution via a couple of variational inference encoders, defined as

$$\mu = \mathcal{F}_\mu^\phi(\mathbf{X}^\phi, \mathbf{A}^\phi), \quad \log(\sigma) = \mathcal{F}_\sigma^\phi(\mathbf{X}^\phi, \mathbf{A}^\phi), \quad (4)$$

where $\mathcal{F}_\mu^\phi$ and $\mathcal{F}_\sigma^\phi$ are variational encoders that map the node features $\mathbf{X}^\phi$ and adjacency matrix $\mathbf{A}^\phi$ to the mean and log-variance of the latent distribution, respectively. Here, μ and σ represent the mean and standard deviation of the latent embeddings. The latent representation $\mathbf{H}^\phi$ is then sampled using the reparameterization trick:

$$\mathbf{H}^\phi = \mu + \epsilon \cdot \sigma, \quad \epsilon \sim \mathcal{N}(0, I). \quad (5)$$

Conditioned on $\mathbf{H}^\phi$, anomalous graphs are generated via dedicated reconstruction heads:

$$\tilde{\mathbf{X}}^\phi = g_x(\mathbf{H}^\phi), \quad \tilde{\mathbf{A}}^\phi = g_a(\mathbf{H}^\phi(\mathbf{H}^\phi)^\top), \quad (6)$$

where g_x and g_a reconstruct node attributes and graph structure, respectively. $\tilde{\mathbf{X}}$ and $\tilde{\mathbf{A}}$ denote the synthesized anomalous node attributes and adjacency matrices. This process enables the generation of graphs with anomalous nodes ($\tilde{\mathbf{X}}^\phi, \mathbf{A}^\phi$), anomalous edges ($\mathbf{X}^\phi, \tilde{\mathbf{A}}^\phi$), and fully anomalous graphs ($\tilde{\mathbf{X}}^\phi, \tilde{\mathbf{A}}^\phi$). The corresponding generative distribution can be expressed as

$$p_\theta(\tilde{\mathcal{G}} | \mathbf{A}, \mathbf{X}) = p_\theta(\tilde{\mathbf{X}} | \mathbf{X})p_\theta(\tilde{\mathbf{A}} | \mathbf{A}^\phi). \quad (7)$$

where p_θ is the prior distribution function. The variational posterior over the latent variable is defined as

$$q_\phi(\mathbf{z} | \mathcal{G}) = \mathcal{N}(\mathbf{z}; \mu_\phi(\mathbf{X}, \mathbf{A}), \Sigma_\phi(\mathbf{X}, \mathbf{A})), \quad (8)$$

with μ_ϕ and Σ_ϕ being neural networks parameterized by ϕ . To regularize the latent space, the distribution of $\mathbf{z}$ is encouraged to approximate a standard Gaussian prior $p(\mathbf{z}) = \mathcal{N}(0, I)$ via the KL divergence:

$$\mathcal{L}_{KL} = \mathbb{E}_{\mathcal{G} \sim p_{data}} [\text{KL}(q_\phi(\mathbf{z} | \mathcal{G}) \| p(\mathbf{z}))]. \quad (9)$$

This latent structure induces a smooth manifold from which both normal and anomalous samples can be drawn. Moreover, to ensure that $\mathcal{G}$ generates meaningful anomalies rather than graphs that resemble normal samples, we introduce an anomaly preservation loss $\mathcal{L}_a$, which encourages a sufficient deviation between generated and normal graphs in the embedding space, as

$$\mathcal{L}_a = \mathbb{E}_{\mathcal{G} \sim p_{data}} [\|\bar{\mu}_\phi(\mathbf{X}, \mathbf{A}) - \bar{\mu}_\phi(\tilde{\mathbf{X}}, \tilde{\mathbf{A}})\|_2^2], \quad (10)$$

where $\bar{\mu}_\phi(\mathbf{X}, \mathbf{A})$ represents the latent mean of the normal graph, and $\bar{\mu}_\phi(\tilde{\mathbf{X}}, \tilde{\mathbf{A}})$ denotes the latent mean of the generated anomalous graph. The generator is optimized with a composite objective:

$$\mathcal{L}_G = \mathcal{L}_{KL} - \mathbb{E}_{\tilde{\mathcal{G}} \sim \mathcal{G}} [\log D(\tilde{\mathcal{G}})] + \mathcal{L}_a, \quad (11)$$

where the discriminator $\mathcal{D}$ feedback (2nd term) is backpropagated to the generator $\mathcal{G}$, guiding it to produce anomalies that are increasingly difficult for the discriminator to distinguish. The improved generative adversarial network effectively synthesizes multi-level anomalous graphs while simultaneously refining the decision

Algorithm 1: Training Procedure of GFM-UAD

Input: Training graph $\mathcal{G} = \{G_i\}_{i=1}^M$ represented as $(\mathbf{X}, \mathbf{A})$;
 Training Epochs N_{epoch} ; Hyper-parameters;
Output: Anomaly score for each node, edge, and graph.

```

1 /* High-confidence Samples Extraction */
2 for  $e \leftarrow 0$  to  $N_{\text{epoch}}$  do
3   Learn the node embeddings by Eq. (1);
4   Extract high-confidence normal samples by Eqs. (2)-(3);
5   Calculate reconstruction loss and backpropagate;
6 end
7 /* Anomaly Samples Generation */
8 for  $e \leftarrow 0$  to  $N_{\text{epoch}}$  do
9   Utilize the reparameterization trick to obtain node
     embedding  $\mathbf{H}^\phi$  from normal samples by Eqs. (4)-(5);
10  Reconstruct node attributes  $\tilde{\mathbf{X}}$  and graph structure  $\tilde{\mathbf{A}}$  to
     generate anomaly samples by Eq. (6);
11  Input generated data in  $\mathcal{D}$  to train by Eq. (12);
12  Obtain detection results by Eqs. (13)-(14);
13  Calculate loss and backpropagate by Eqs. (11) and (16);
14 end
15 /* Domain-driven Fine-tuning */
16 Fine-tune the detector on the target dataset by Eq. (16);
17 Perform multi-level anomaly detection by Eqs. (13)-(14);
18 return  $s^{\mathcal{G}}$ ,  $s^n$  and  $s^e$ .
```

boundary of the discriminator $\mathcal{D}$, improving the capability of the model to detect previously unseen anomaly patterns.

Multi-Level Discriminator via Message Passing To effectively distinguish normal instances from generated anomalies at multiple levels, we design a multi-level discriminator $\mathcal{D}$ that operates on node, edge, and graph-level representations through a message passing mechanism. This allows the detector to capture both local and global inconsistencies introduced by the generator. At each level, we leverage forward representation and score back-propagation (i.e., the anomaly score of the next level is spliced into the representation of the previous level). Specifically, the discriminator $\mathcal{D}$ can be formal as $\mathcal{D}_h \circ \mathcal{D}_e$, where $\mathcal{D}_e$ is projector and $\mathcal{D}_h$ is the detector head. For Node-level discrimination, we utilize $\mathcal{D}_e$ to extract node-level embeddings, as

$$\mathbf{h}_i^{(l+1)} = \sigma(\mathbf{W}_n^{(l)} \mathbf{h}_i^{(l)} + \sum_{j \in \mathcal{N}(i)} \frac{1}{|\mathcal{N}(i)|} \mathbf{W}_n^{(l)} \mathbf{h}_j^{(l)}), \quad (12)$$

where $\mathbf{h}_i^{(0)} = \mathbf{x}_i$ is the input node feature, $\mathcal{N}(i)$ denotes the set of neighbors of node i , $\mathbf{W}_n^{(l)}$ is a learnable weight matrix at layer l , and σ is a nonlinear activation. The node-level anomaly score is computed as:

$$s_i^n = f_n(\mathbf{h}_i^{(L)}, \text{agg}(\mathcal{N}(i))), \quad (13)$$

where $\mathcal{N}(i)$ denotes the neighborhood of node i , and $\text{agg}(\mathcal{N}(i))$ is an aggregation function that combines the neighborhood information of node i . The function f_n is an MLP that takes as input the node embedding $\mathbf{h}_i^{(L)}$ at the final layer L , along with the aggregated neighborhood information $\text{agg}(\mathcal{N}(i))$. The MLP outputs

a likelihood score, s_i^n , representing the probability of node i being anomalous. Then, for edge-level discrimination, we define an edge (i, j) embedding using the concatenation of connected node embeddings and propagate edge messages:

$$\mathbf{e}_{ij}^{(l+1)} = \sigma(\mathbf{W}_e^{(l)} \mathbf{e}_{ij}^{(l)} + \mathbf{W}_h^{(l)} (\mathbf{h}_i^{(l)} \parallel \mathbf{h}_j^{(l)})), \quad (14)$$

where $\mathbf{e}_{ij}^{(0)}$ be initialized as the adjacency entry $\mathbf{A}_{ij}$ by random walk, $\parallel$ denotes concatenation, and $\mathbf{W}_e^{(l)}, \mathbf{W}_h^{(l)}$ are learnable parameters. The edge-level anomaly score is $s_{ij}^e = f_e(\mathbf{e}_{ij}^{(L)}, \text{agg}(\mathcal{N}_e(i, j)))$, with f_e as a simple MLP predicting edge abnormality likelihood. $\text{agg}(\mathcal{N}(i))$ is an aggregation function that combines the neighborhood information of edge ij . About graph-level discrimination, the graph-level representation is obtained by aggregating node embeddings with a readout function: $\mathbf{h}_{\mathcal{G}} = \text{READOUT}(\{\mathbf{h}_i^{(L)} \mid i \in \mathcal{V}\} \cup \sum_{(i,j) \in E} w_{ij} \mathbf{e}_{ij}^{(L)})$, where READOUT can be mean, sum pooling. The overall graph anomaly score is $s_{\mathcal{G}}^{\mathcal{G}} = f_g(\mathbf{h}_{\mathcal{G}}, \text{agg}(\mathcal{N}_{\mathcal{G}}(i)))$, where f_g is a detector head, which outputs the probability of the graph being anomalous. Finally, the multi-level discriminator is trained to maximize the likelihood of correctly classifying normal and generated multi-level objectives:

$$\mathcal{L}_D = -\mathbb{E}_{\mathcal{G} \sim p_{\text{data}}} [\log D(\mathcal{G})] - \mathbb{E}_{\tilde{\mathcal{G}} \sim G} [\log(1 - D(\tilde{\mathcal{G}}))], \quad (15)$$

where $D(\cdot)$ integrates the scores at the node, edge, and graph level, as $D(\mathcal{G}) = \frac{1}{|\mathcal{V}|} \sum_i s_i^n + \frac{1}{|\mathcal{E}|} \sum_{(i,j) \in E} s_{ij}^e + s_{\mathcal{G}}^{\mathcal{G}}$. This design allows the discriminator to leverage hierarchical information propagated through message passing to robustly detect anomalies at different scales. In this network, adversarial learning is leveraged to synthesize diverse and informative anomalous instances. By doing so, the model overcomes the limitations of conventional dataset-specific training and improves its generalization across different anomaly types and environments to detect previously unseen anomalies.

3.4 Domain-Driven Fine-Tuning

In this stage, we fine-tune the pretrained multi-level anomaly detection head to adapt to the specific environment. Unlike traditional supervised methods, our approach leverages the inherent structure of the latent representations and the normal sample distribution to prompt the detection head, dynamically adjusting the anomaly boundaries, in an unsupervised manner.

Specifically, we first employ K -means to identify the larger cluster as normal, then select the T samples closest to its centroid target dataset. Next, we obtain these samples node-level embeddings from the pretrained encoder of the discriminator. We then consider these embeddings collectively to define the empirical distribution of normal representations in the target dataset, denoted as $q_{\text{target}}(\mathbf{h})$. Similarly, the pretrained model provides a reference distribution $q_{\text{pretrain}}(\mathbf{h})$ learned from the source dataset. The detection head is then fine-tuned by aligning the target distribution with the pretraining distribution in the embedding space. This can be formulated as minimizing a distributional divergence:

$$\mathcal{L}_{\text{align}} = JS(q_{\text{target}}(\mathbf{h}) \parallel q_{\text{pretrain}}(\mathbf{h})), \quad (16)$$

where $JS(\cdot \parallel \cdot)$ denote Jensen-Shannon divergence. This encourages the detection head to adapt to the target domain while preserving the structural properties learned during pre-training. In the same

Table 1: Statistics of the single graph (upper part) and multiple graph (lower part) datasets. A.Nodes, A.Numms denote the average number of nodes and average node numbers, respectively. The ratio denotes the anomaly sample ratio.

Dataset	Nodes	Edges	Features	Ratio
Amazon	13,752	515,042	767	5.0%
Reddit	10,948	168,016	64	3.3%
Weibo	8,405	407,963	400	10.3%
YelpChi	24,741	49,315	597	2.4%
Amazon-all	10,000	200,000	500	4.1%
YelpChi-all	45,941	3,876,979	32	14.52%
Disney	124	335	28	4.8%
T-Finance	39,357	21,222,543	10	4.6%
T-Social	5,781,065	73,105,508	10	3.0%

Datasets	Graphs	A.Nodes	A.Numms	Ratio
MUTAG	188	67.1	30.32	35.7%
BZR	405	86	35.75	21.23%
COX2	467	98	41.22	21.84%
DHFR	756	295	42.43	39.02%
ENZYMES	600	100	32.63	16.67%
PROTEINS	1113	450	39.06	40.43%
DD	1178	487	284.32	41.34%
SW-620	40532	2420	26.06	5.95%
PC-3	27509	1568	26.36	4.30%

way, we synchronize to achieve the edge-level and graph-level distribution alignment.

3.5 Complexity Analysis

The computational complexity of the GFM-UAD approach can be broken down into several stages, each contributing differently based on the operations involved. In the high-confidence sample extraction stage, the main computational cost arises from passing each graph through the Graph Neural Network (GNN) encoder with L_{GNN} layers, which has a complexity of $O(L_{\text{GNN}}(|E|d + |V|d^2))$, $|E|$ is the number of edges, $|V|$ is the number of nodes. For simplicity in computation, the dimensions of the hidden layers and the original attributes are both considered to be d . In the pre-training stage, the variational encoders and decoders require $O(L_{\text{VAE}}(|E|d + |V|d^2))$ operations per graph, where L_{VAE} is the number of layers in the graph variational autoencoder. Finally, in the Domain-Driven Fine-Tuning stage, the distributional alignment through Jensen-Shannon divergence adds a complexity of $O(N_s d |V|)$, where N_s is the number of graphs. In summary, the complexity of this method is primarily determined by $O((L_{\text{GNN}} + L_{\text{VAE}}) \times (|E|d + |V|d^2))$. Therefore, the complexity of this framework exhibits only a linear growth in computational complexity compared to commonly used GCN methods, demonstrating its scalability for large-scale datasets.

4 Experiments

In this section, we conduct a comprehensive evaluation of our proposed method. We first present comparison experiments against state-of-the-art baselines, followed by ablation studies to analyze the contribution of each component, and hyperparameter sensitivity analysis to investigate the robustness of our approach under

various settings. Moreover, we further perform convergence experiments to validate the stability and effectiveness of GFM-UAD.

4.1 Experiential Settings

4.1.1 Evaluation Metrics. We adopt the Area Under the Receiver Operating Characteristic Curve (AU-ROC) and the Area Under the Precision-Recall Curve (AU-PRC) as the primary evaluation metrics [40, 58]. The details of these metrics are shown in **Appendix B**.

4.1.2 Baseline Methods. To comprehensively evaluate our proposed approach, we compare it with baseline methods across two categories and three-level tasks: the categories include (1) advanced deep learning methods and (2) graph foundation models, while the levels encompass node-level, graph-level, and edge-level anomaly detection tasks. Specifically, advanced deep learning methods consist of node-level methods, including AnomalyDAE [10], CoLA [35], CONAD [52], HCM-A [18], TAM [41], GADAM [6], CARE [62]; graph-level methods, including iGAD [59], GLocalKD [37], GOOD-D [33], HC-GLAD [11], MUSE [21], SDGG [1]; edge-level methods, including BOURNE [30], RGAE [36]; unified anomaly detection methods, including Uniform [45] and UniGAD [28]. The GFM models include AnomalyGFM [40], UNPrompt [39], and ARC [34]. The details of these baseline methods are shown in **Appendix C**.

4.1.3 Benchmark Datasets. To evaluate the generalization capability of UFM-UAD, we employ two categories (single graph and multiple graphs) of datasets. The single graph datasets perform node-level and edge-level tasks, including Amazon, Reddit, Weibo, YelpChi, Amazon-all, YelpChi-all, and Disney [3, 44], and multiple graph datasets perform graph-level and node-level tasks, including MUTAG, BZR, COX2, DHFR, PROTEINS, IMDB, and DD. Furthermore, to ensure that GFM-UAD maintains consistent generalization capability on large datasets, we selected T-Finance and T-Social as single graph datasets, and PC-3 and SW-620 as multi-graph datasets. The detailed information of these benchmark datasets is shown in Table 1 and **Appendix D**.

4.1.4 Implementation Details. For a fair comparison, each experiment is independently repeated five times. We use the Adam optimizer with a learning rate of 0.001, which decays at a rate of $5e - 4$, and a batch size of 256. The model is trained for 100 epochs on all datasets. For additional implementation details, see **Appendix E**.

4.2 Comparison Experiments

To validate the effectiveness of GFM-UAD, we designed two sets of joint tasks: node-level and edge-level tasks on single-graph datasets and node-level and graph-level tasks on multi-graph datasets. The experimental results for node-level tasks on the single graph datasets and graph-level tasks on the multiple graph datasets are presented in Table 2. Other results are detailed in **Appendix F.1**. From the results, we observe the following conclusions: 1) Compared with SOTA deep anomaly detection methods, our method shows significant performance improvements on node-level, edge-level, and graph-level tasks. This is primarily due to the fact that existing deep models are designed solely for the current domain knowledge during training, lacking the ability to generalize across domains. 2) Compared to advanced GFM models, our method still demonstrates significant performance improvement. This is primarily due to the

Table 2: AUROC and AUPRC results for node-level tasks and graph-level tasks, using models trained on Facebook and AIDS datasets only. For each dataset, the best performance within each metric is **bolded, and the second-best is underlined. “OOT” and “OOM” represent timeout and memory limit exceeded, respectively.**

Node-level Tasks on Single Graph Datasets																			
Methods	Amazon		Reddit		Weibo		YelpChi		Aamzon-all		YelpChi-all		Disney		T-Finance		T-Social		
	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	
AnomalyDAE	58.2 $\pm$ 3.9	8.3 $\pm$ 1.5	50.2 $\pm$ 3.2	3.2 $\pm$ 0.4	77.8 $\pm$ 5.8	60.6 $\pm$ 3.1	48.3 $\pm$ 9.4	6.4 $\pm$ 1.7	72.3 $\pm$ 2.3	19.2 $\pm$ 2.6	50.0 $\pm$ 1.8	14.8 $\pm$ 0.9	48.4 $\pm$ 0.3	5.6 $\pm$ 0.0	40.1 $\pm$ 1.7	3.5 $\pm$ 2.2	23.8 $\pm$ 0.9	5.4 $\pm$ 1.4	
CoLA	45.8 $\pm$ 5.4	6.7 $\pm$ 0.2	46.2 $\pm$ 0.5	3.9 $\pm$ 0.4	39.2 $\pm$ 4.1	11.8 $\pm$ 0.1	49.1 $\pm$ 1.7	5.1 $\pm$ 0.0	40.9 $\pm$ 5.2	8.7 $\pm$ 1.9	48.8 $\pm$ 1.0	14.6 $\pm$ 0.3	46.9 $\pm$ 6.5	7.1 $\pm$ 2.3	38.5 $\pm$ 1.2	5.6 $\pm$ 2.5	28.1 $\pm$ 3.4	6.5 $\pm$ 2.3	
HCM-A	47.8 $\pm$ 0.5	6.7 $\pm$ 0.2	53.9 $\pm$ 4.1	3.9 $\pm$ 0.4	57.8 $\pm$ 4.8	11.8 $\pm$ 0.1	50.0 $\pm$ 0.0	5.1 $\pm$ 0.0	50.6 $\pm$ 5.9	8.1 $\pm$ 1.9	50.2 $\pm$ 0.5	14.6 $\pm$ 0.3	20.1 $\pm$ 1.5	3.5 $\pm$ 0.1	41.9 $\pm$ 1.4	4.2 $\pm$ 1.6	24.6 $\pm$ 1.3	4.9 $\pm$ 1.4	
TAM	47.2 $\pm$ 0.5	6.7 $\pm$ 0.1	57.2 $\pm$ 0.4	4.1 $\pm$ 0.1	48.7 $\pm$ 2.8	12.4 $\pm$ 1.4	50.4 $\pm$ 1.4	5.2 $\pm$ 0.2	75.4 $\pm$ 0.2	17.3 $\pm$ 0.4	42.2 $\pm$ 0.2	12.4 $\pm$ 0.1	47.7 $\pm$ 0.3	6.2 $\pm$ 0.2	42.3 $\pm$ 2.5	6.2 $\pm$ 0.5	38.4 $\pm$ 1.6	4.4 $\pm$ 1.9	
GADAM	66.5 $\pm$ 6.3	15.6 $\pm$ 1.3	45.3 $\pm$ 2.4	2.9 $\pm$ 0.1	36.5 $\pm$ 5.2	8.3 $\pm$ 0.5	33.7 $\pm$ 1.2	3.5 $\pm$ 0.1	59.6 $\pm$ 0.8	10.3 $\pm$ 12.4	48.2 $\pm$ 1.6	13.7 $\pm$ 0.6	42.9 $\pm$ 2.3	6.5 $\pm$ 0.1	OOM	OOM	OOM	OOM	
CARE	68.4 $\pm$ 1.5	20.2 $\pm$ 0.6	40.7 $\pm$ 0.5	2.5 $\pm$ 1.3	77.9 $\pm$ 1.3	61.4 $\pm$ 0.5	OOM	OOM	OOT	OOT	OOM	OOM	59.8 $\pm$ 0.6	7.8 $\pm$ 1.5	OOT	OOT	OOT	OOT	
UniGAD	51.5 $\pm$ 1.3	7.2 $\pm$ 3.0	54.6 $\pm$ 1.4	4.4 $\pm$ 0.2	48.3 $\pm$ 0.6	13.1 $\pm$ 0.3	55.6 $\pm$ 1.8	4.5 $\pm$ 1.4	64.4 $\pm$ 1.2	8.3 $\pm$ 2.4	51.3 $\pm$ 2.0	14.1 $\pm$ 1.2	60.2 $\pm$ 1.4	9.2 $\pm$ 2.3	OOM	OOM	OOM	OOM	
Uniform	46.3 $\pm$ 2.4	5.5 $\pm$ 0.9	51.1 $\pm$ 0.6	3.6 $\pm$ 0.2	45.7 $\pm$ 0.3	10.5 $\pm$ 1.4	53.5 $\pm$ 0.9	4.7 $\pm$ 1.6	47.3 $\pm$ 0.2	3.5 $\pm$ 1.0	49.4 $\pm$ 0.7	14.0 $\pm$ 0.0	35.6 $\pm$ 1.4	4.8 $\pm$ 0.1	42.7 $\pm$ 0.3	4.5 $\pm$ 0.2	35.6 $\pm$ 1.2	3.6 $\pm$ 0.7	
UNPrompt	60.1 $\pm$ 8.3	13.6 $\pm$ 1.6	50.8 $\pm$ 0.3	3.2 $\pm$ 0.7	76.4 $\pm$ 0.7	60.8 $\pm$ 1.2	52.3 $\pm$ 0.5	5.5 $\pm$ 1.6	63.1 $\pm$ 1.4	15.2 $\pm$ 0.7	50.9 $\pm$ 1.8	14.3 $\pm$ 0.7	49.7 $\pm$ 3.7	6.0 $\pm$ 1.0	19.9 $\pm$ 0.8	2.6 $\pm$ 1.2	34.3 $\pm$ 1.4	3.5 $\pm$ 1.9	
ARC	63.6 $\pm$ 2.2	14.2 $\pm$ 1.2	59.7 $\pm$ 0.4	9.9 $\pm$ 1.3	78.5 $\pm$ 1.3	62.4 $\pm$ 0.6	48.5 $\pm$ 0.5	5.9 $\pm$ 0.1	34.9 $\pm$ 2.9	4.8 $\pm$ 0.2	57.6 $\pm$ 2.7	16.4 $\pm$ 1.6	50.1 $\pm$ 0.3	4.9 $\pm$ 0.8	19.8 $\pm$ 1.7	2.5 $\pm$ 0.0	OOM	OOM	
AnomalyGFM	68.1 $\pm$ 1.4	21.2 $\pm$ 0.6	54.3 $\pm$ 1.8	3.5 $\pm$ 1.5	59.2 $\pm$ 1.6	18.6 $\pm$ 0.5	55.1 $\pm$ 1.4	6.6 $\pm$ 3.7	70.8 $\pm$ 1.9	16.6 $\pm$ 1.3	60.3 $\pm$ 2.5	17.2 $\pm$ 2.0	55.3 $\pm$ 1.6	8.9 $\pm$ 1.5	37.5 $\pm$ 1.2	5.4 $\pm$ 1.3	40.2 $\pm$ 1.0	4.3 $\pm$ 0.9	
OURS	72.2 $\pm$ 1.5	23.6 $\pm$ 0.7	62.8 $\pm$ 1.6	12.6 $\pm$ 1.2	81.5 $\pm$ 3.4	63.2 $\pm$ 1.3	59.4 $\pm$ 1.2	8.8 $\pm$ 1.3	75.4 $\pm$ 3.2	23.7 $\pm$ 2.2	64.7 $\pm$ 1.5	19.5 $\pm$ 1.0	67.2 $\pm$ 1.6	9.5 $\pm$ 0.7	45.5 $\pm$ 0.6	7.7 $\pm$ 0.9	42.3 $\pm$ 1.2	5.5 $\pm$ 1.5	
Graph-level Tasks on Multiple Graph Datasets																			
	MUTAG		BZR		COX2		DHFR		ENZYMES		PROTEINS		DD		SW-620		PC-3		
GLocalKD	12.8 $\pm$ 2.4	47.6 $\pm$ 3.5	44.2 $\pm$ 2.1	67.3 $\pm$ 3.5	36.9 $\pm$ 3.4	78.2 $\pm$ 1.3	48.8 $\pm$ 1.7	62.5 $\pm$ 2.6	49.3 $\pm$ 3.8	17.2 $\pm$ 2.6	70.1 $\pm$ 2.4	70.3 $\pm$ 2.5	OOM	OOM	34.7 $\pm$ 2.1	4.8 $\pm$ 1.6	32.6 $\pm$ 3.1	3.5 $\pm$ 2.9	
GOOD-D	41.7 $\pm$ 2.6	43.5 $\pm$ 1.8	38.6 $\pm$ 3.5	55.6 $\pm$ 2.7	40.5 $\pm$ 3.5	60.8 $\pm$ 2.7	46.4 $\pm$ 2.9	50.3 $\pm$ 2.3	62.2 $\pm$ 3.5	18.9 $\pm$ 4.7	68.8 $\pm$ 4.7	47.6 $\pm$ 4.7	OOM	OOM	OOM	OOM	OOM	OOM	
MUSE	53.4 $\pm$ 5.8	64.9 $\pm$ 2.6	57.1 $\pm$ 18.1	54.2 $\pm$ 15.2	57.6 $\pm$ 7.9	51.2 $\pm$ 8.4	57.8 $\pm$ 8.2	63.5 $\pm$ 5.9	58.6 $\pm$ 9.7	22.4 $\pm$ 3.6	59.6 $\pm$ 3.8	15.9 $\pm$ 12.1	72.5 $\pm$ 4.1	70.5 $\pm$ 5.6	72.6 $\pm$ 5.5	11.6 $\pm$ 2.4	OOM	OOM	
iGAD	54.3 $\pm$ 5.1	45.2 $\pm$ 1.5	48.0 $\pm$ 7.6	21.9 $\pm$ 1.9	56.6 $\pm$ 6.1	22.3 $\pm$ 0.0	49.2 $\pm$ 2.9	61.2 $\pm$ 0.0	49.2 $\pm$ 2.9	61.3 $\pm$ 0.7	71.3 $\pm$ 1.8	49.5 $\pm$ 1.6	OOM	OOM	45.2 $\pm$ 6.9	5.9 $\pm$ 0.0	48.1 $\pm$ 6.3	5.6 $\pm$ 1.5	
HC-GLAD	5.7 $\pm$ 0.7	21.5 $\pm$ 1.6	51.7 $\pm$ 6.4	68.8 $\pm$ 3.5	42.6 $\pm$ 5.7	63.7 $\pm$ 5.6	OOM	OOM	60.5 $\pm$ 2.1	30.4 $\pm$ 1.2	66.7 $\pm$ 1.5	67.2 $\pm$ 2.1	OOM	OOM	OOM	OOM	OOM	OOM	
SDGG	51.4 $\pm$ 6.3	44.8 $\pm$ 3.9	55.4 $\pm$ 2.9	26.7 $\pm$ 2.4	61.4 $\pm$ 3.2	65.5 $\pm$ 2.1	53.8 $\pm$ 2.6	60.2 $\pm$ 1.5	59.3 $\pm$ 1.6	25.4 $\pm$ 1.3	68.5 $\pm$ 3.2	59.7 $\pm$ 2.5	71.3 $\pm$ 4.9	68.3 $\pm$ 5.6	59.4 $\pm$ 7.1	8.5 $\pm$ 2.9	44.3 $\pm$ 1.7	4.2 $\pm$ 1.5	
UniGAD	58.1 $\pm$ 0.4	43.5 $\pm$ 1.9	49.6 $\pm$ 3.5	53.1 $\pm$ 1.4	48.6 $\pm$ 1.5	29.6 $\pm$ 2.4	55.2 $\pm$ 2.3	27.6 $\pm$ 1.0	61.2 $\pm$ 1.5	32.3 $\pm$ 2.4	58.4 $\pm$ 3.2	21.3 $\pm$ 4.7	56.4 $\pm$ 3.5	47.3 $\pm$ 2.8	51.6 $\pm$ 1.2	6.7 $\pm$ 1.1	54.9 $\pm$ 4.5	8.5 $\pm$ 3.2	
Uniform	62.5 $\pm$ 1.5	41.6 $\pm$ 2.2	67.3 $\pm$ 2.7	64.2 $\pm$ 2.5	60.7 $\pm$ 1.3	58.7 $\pm$ 1.3	57.5 $\pm$ 2.1	48.7 $\pm$ 1.4	54.7 $\pm$ 2.5	24.5 $\pm$ 1.3	64.3 $\pm$ 3.5	59.5 $\pm$ 2.4	OOM	OOM	OOM	OOM	OOM	OOM	
UNPrompt	47.5 $\pm$ 1.4	38.6 $\pm$ 1.9	47.2 $\pm$ 1.3	34.8 $\pm$ 2.1	61.9 $\pm$ 3.4	37.6 $\pm$ 1.6	68.7 $\pm$ 2.4	64.7 $\pm$ 2.3	46.9 $\pm$ 1.7	34.2 $\pm$ 1.6	65.5 $\pm$ 2.3	43.2 $\pm$ 1.6	OOM	OOM	OOM	OOM	OOM	OOM	
ARC	50.5 $\pm$ 1.6	49.6 $\pm$ 2.8	57.9 $\pm$ 2.3	50.4 $\pm$ 1.8	59.7 $\pm$ 3.4	57.4 $\pm$ 1.9	62.8 $\pm$ 2.1	63.5 $\pm$ 2.1	55.6 $\pm$ 1.8	27.4 $\pm$ 1.7	72.2 $\pm$ 1.5	60.3 $\pm$ 1.3	68.5 $\pm$ 1.6	64.4 $\pm$ 1.7	OOM	OOM	OOM	OOM	
AnomalyGFM	57.7 $\pm$ 3.4	53.6 $\pm$ 3.0	62.1 $\pm$ 3.5	58.6 $\pm$ 2.2	68.2 $\pm$ 3.4	54.7 $\pm$ 3.2	48.9 $\pm$ 1.6	32.8 $\pm$ 1.4	57.3 $\pm$ 2.2	28.4 $\pm$ 1.7	71.6 $\pm$ 1.7	69.0 $\pm$ 1.4	69.2 $\pm$ 1.4	66.3 $\pm$ 1.3	OOM	OOM	OOM	OOM	
OURS	67.8 $\pm$ 2.1	68.6 $\pm$ 2.4	73.5 $\pm$ 1.9	72.4 $\pm$ 1.8	74.2 $\pm$ 1.7	80.1 $\pm$ 3.2	70.1 $\pm$ 2.1	68.4 $\pm$ 1.1	74.7 $\pm$ 1.4	75.6 $\pm$ 1.1	73.5 $\pm$ 2.6	71.2 $\pm$ 2.8	73.4 $\pm$ 2.1	71.1 $\pm$ 1.5	74.7 $\pm$ 1.3	13.5 $\pm$ 1.2	56.8 $\pm$ 0.7	7.2 $\pm$ 0.6	

Table 3: Performance of GFM-UAD and its four variants for node-level tasks and graph-level tasks.

Variants	Metrics	Amazon	Reddit	Weibo	BZR	COX2	DHFR
None	AUROC	48.9 $\pm$ 1.4	47.6 $\pm$ 0.9	47.3 $\pm$ 2.0	42.1 $\pm$ 1.8	36.9 $\pm$ 1.2	50.3 $\pm$ 1.6
	AUPRC	5.7 $\pm$ 1.2	3.1 $\pm$ 1.6	7.7 $\pm$ 0.5	28.7 $\pm$ 2.5	37.4 $\pm$ 0.8	52.5 $\pm$ 0.6
w/o AG	AUROC	53.3 $\pm$ 1.2	49.2 $\pm$ 3.5	56.8 $\pm$ 2.0	52.6 $\pm$ 1.4	43.9 $\pm$ 3.1	52.6 $\pm$ 1.8
	AUPRC	4.6 $\pm$ 1.2	3.8 $\pm$ 1.5	12.4 $\pm$ 1.3	38.5 $\pm$ 1.9	52.5 $\pm$ 2.0	54.7 $\pm$ 1.5
w/o MF	AUROC	71.4 $\pm$ 2.1	57.7 $\pm$ 1.4	74.3 $\pm$ 3.1	62.4 $\pm$ 2.1	72.1 $\pm$ 1.6	68.3 $\pm$ 1.4
	AUPRC	16.5 $\pm$ 1.7	8.7 $\pm$ 1.4	58.6 $\pm$ 1.3	60.1 $\pm$ 1.2	67.7 $\pm$ 1.5	60.1 $\pm$ 2.0
w/o MIP	AUROC	70.3 $\pm$ 1.6	59.6 $\pm$ 1.5	78.6 $\pm$ 2.6	63.9 $\pm$ 1.5	68.6 $\pm$ 1.3	64.8 $\pm$ 2.0
	AUPRC	20.1 $\pm$ 0.5	11.5 $\pm$ 1.5	61.4 $\pm$ 1.5	70.5 $\pm$ 1.5	75.5 $\pm$ 1.8	63.2 $\pm$ 2.5
OURS	AUROC	72.2 $\pm$ 1.5	62.8 $\pm$ 1.6	81.5 $\pm$ 3.4	73.5 $\pm$ 1.9	74.2 $\pm$ 1.7	70.1 $\pm$ 2.1
	AUPRC	23.6 $\pm$ 0.7	12.6 $\pm$ 1.2	63.2 $\pm$ 1.3	72.4 $\pm$ 1.8	80.1 $\pm$ 3.2	68.4 $\pm$ 1.1

fact that these methods heavily rely on label guidance in the pre-training stage. Once the labels are lost, the parameters of these models cannot be properly optimized, which hinders the generation of high-quality representations. The fine-tuning process without label guidance fails to leverage the prior knowledge necessary to distinguish anomaly boundaries. 3) Moreover, on large-scale datasets, our method still demonstrates advantages, which can be mainly attributed to the lightweight nature of our method’s components compared to SOTA approaches. At the same time, through the process of proactive generalization, our method has acquired

sufficient knowledge during the pre-training phase, allowing high-quality detection in the inference phase without requiring complex computational processes. 4) We observe that there is a significant difference between AUROC and AUPRC across different datasets. This is mainly due to the inherent characteristics of the datasets, particularly their balance. However, GFM-UAD still shows a significant advantage in handling imbalanced samples, which can be attributed to the fact that our model is able to cover these subtle patterns that remain undetected and unrecognized through the process of proactive generalization.

4.3 Generalization Ability Verification

To ensure consistent and reliable performance across diverse domains, we perform pre-training on several benchmark datasets. This process validates the effectiveness of GFM-UAD in adapting to various environments. The experimental results are shown in Fig. 3, and the following conclusions can be drawn: 1) No matter which dataset is used for pre-training, our proposed method still significantly outperforms existing SOTA approaches. This is attributed to its strong capability for domain anomaly generalization and knowledge transfer. 2) Compared with node-level tasks, graph-level tasks provide better generalization capability. This is mainly because graph-level datasets contain richer and complete structure information, enabling the detector to capture associations from

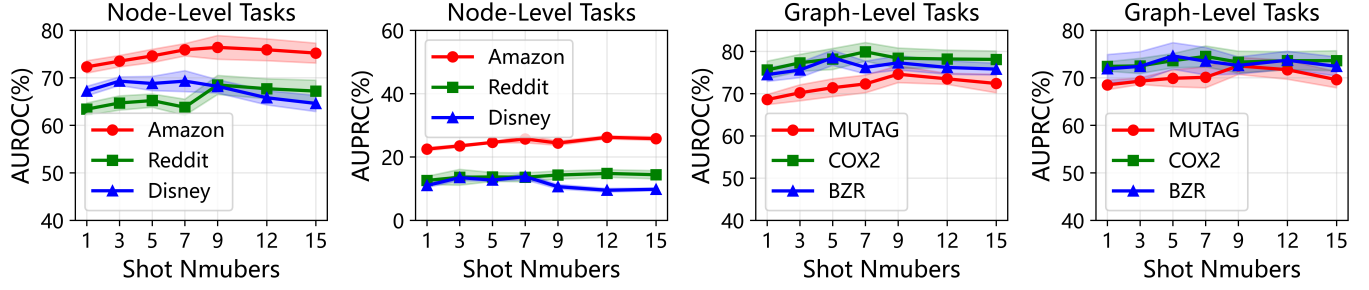


Figure 2: AUROC and AUPRC performance of GFM-UAD under different numbers of shots on six benchmark datasets.

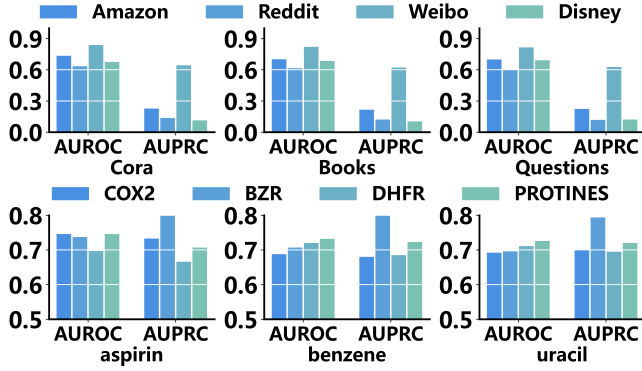


Figure 3: Generalization performance under pre-training on different datasets (shown on “title”) for node- and graph-level tasks on the single and multiple graph datasets, respectively.

multiple perspectives for identifying unseen samples. 3) The final performance variations caused by pre-training on different datasets can be attributed to the differences in the inherent distribution patterns of the datasets. When the patterns of normal samples in some datasets closely resemble the patterns of anomalous samples in others, overall performance tends to degrade. The fine-tuning process helps to some extent in correcting these false positives. 4) In fact, we can pre-train on mixed datasets to enable better generalization.

4.4 Ablation Studies

To validate the effectiveness of each component in the proposed anomaly generation strategy and multi-level fine-tune mechanism, we conduct a series of ablation studies. Specifically, we design several variants of our model by removing or modifying key components: (1) without anomaly generation (**w/o AG**), where anomalous graphs are not synthesized from normal samples; Instead, we directly use the abnormal samples from the previous stage to replace. (2) without multi-level fine-tune (**w/o MF**), where the fine-tune module is excluded and only the initial parameters are used; (3) without the multi-level information passing (**w/o MIP**) mechanism on the detection head; (4) full model, which incorporates both anomaly generation and multi-level refinement. By comparing the performance of these variants with our complete framework, we can quantitatively assess the contribution of each component to the overall anomaly detection performance. 1) Removing the heuristic

anomaly generation mechanism, the model’s performance drops significantly. This degradation is mainly attributed to the fact that the detector trained solely on the original dataset lacks sufficient generalization ability, as the parameters cannot adapt well to diverse anomalous patterns. 2) Removing the MF, the performance of GFM-UAD to detect anomalous samples remains at a relatively high level, primarily due to the guidance paradigm for anomalies being actively generated. In contrast, many normal samples are misclassified as anomalies, which can be mainly attributed to the absence of a normal distribution correction model in the detection head, leading to biased decision boundaries. 3) Removing the message-passing detection head, the accuracy of node- and graph-level anomaly detection will decrease because the key discriminative information cannot be shared, blurring the boundaries of anomaly samples.

4.5 Hyper-parameter Sensitivity Analysis

4.5.1 Generation Indicator θ and ζ . θ and ζ control the number of high-confident normal samples and the anomaly sample generation rate, respectively. To evaluate the influence of the two parameters on the overall performance, we vary their values and observe the resulting changes in key metrics. This analysis enables us to quantitatively assess the sensitivity of the model to varying levels of generated anomaly samples and to determine the optimal settings of ζ and θ that maximize detection effectiveness. The experimental results are presented in Fig. 4, from which we draw the following conclusions: 1) The selection of θ at 10% is most advantageous for performance. A value that is too small may lead to an insufficient number of normal samples, ultimately resulting in false positives. Conversely, too large θ may blur the boundary between normal and anomalous samples, thereby reducing the quality of the generated anomalies. 2) The performance varies with different choices of ζ and θ . When θ is small, ζ should also be small, as this enhances anomaly detection accuracy, although it may reduce the generalization ability of the model. On the other hand, when both θ and ζ are large, the method exhibits better performance on complex datasets. Moreover, if computational resources permit, a larger value of ζ is generally more beneficial for improving performance.

4.5.2 Fine-tune Number of Samples T . To analyze the performance of our proposed method under different fine-tuning settings, we present the results by varying the shot numbers, as shown in Fig. 2. Additional experimental results are shown in **Appendix F.2**. The analysis of the results reveals several important findings: Increasing

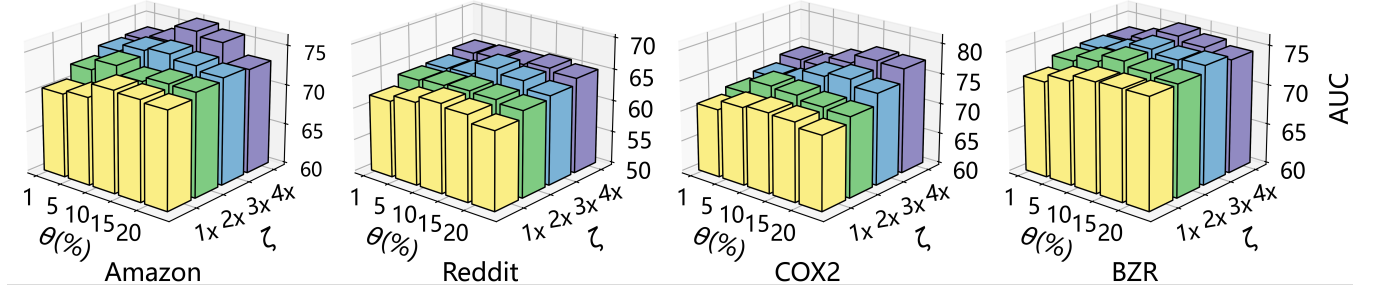


Figure 4: AUROC performance for node-level on single datasets (Amazon and Reddit) and graph-level tasks on multiple graph datasets (COX2 and BZR) of GFM-UAD under varying ratios of θ and ζ .

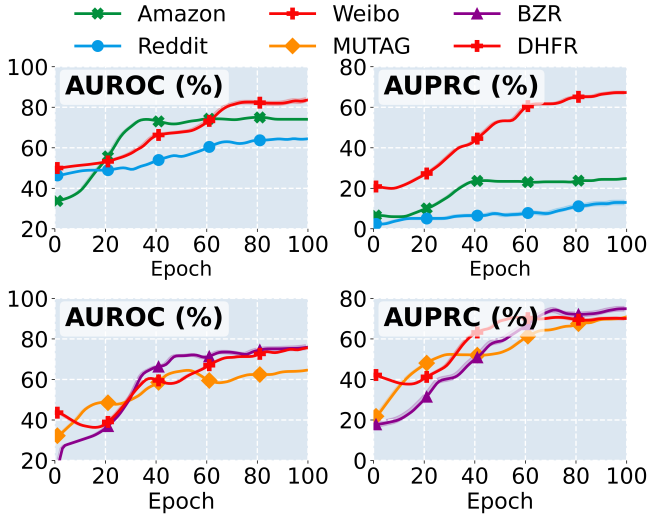


Figure 5: Convergence curve of the GFM-UAD for node-level (first row) and graph-level (second row) tasks over 100 training epochs, reporting AUROC and AUPRC metrics.

the number of fine-tuning samples leads to improved model performance; However, after reaching a certain threshold, the performance improvement gradually decreases as the number continues to increase. The threshold is related to the inherent characteristics of different datasets. This phenomenon is primarily attributed to the fact that only representative samples can effectively distill knowledge to adapt to the target domain.

4.6 Convergence Analysis

We examine the convergence behavior of GFM-UAD during training on six datasets. As shown in Fig. 5. Both performance metrics exhibit a steady improvement across training epochs, indicating a smooth and consistent progression. Furthermore, we observe that the performance improvement for node-level tasks is more stable compared to that for graph-level tasks. This can primarily be attributed to the fact that node-level tasks are generally more straightforward compared to graph-level tasks.

Table 4: Wall-clock training time per epoch (seconds) on large-scale datasets.

Method	Pre-train	Inference	
		T-Finance	T-Social
ARC	13.2	27.4	65.3
AnomalyGFM	15.3	28.7	39.8
GFM-UAD	8.6	12.5	32.6

4.7 Wall-clock Time Analysis

To evaluate computational efficiency, we conduct wall-clock time experiments on two large-scale datasets, comparing GFM-UAD with ARC and AnomalyGFM in both pre-training and inference. As shown in Table 4, GFM-UAD consistently achieves the best efficiency across all settings, with lower training time per epoch and faster inference on both medium- and large-scale graphs. This advantage is particularly pronounced on the larger T-Social dataset, demonstrating better scalability with respect to graph size. The efficiency gains mainly arise from a unified anomaly detection design guided by an active generalization mechanism. This design avoids unnecessary architectural complexity and computational overhead, enabling a favorable trade-off between detection performance and runtime efficiency.

5 Conclusion

In this paper, we propose a unified graph foundation model to address the anomaly detection task under an unsupervised scenario. Compared with existing state-of-the-art GFM-based approaches for GAD, our method, GFM-UAD, leverages advanced heuristic anomaly generation and multi-level knowledge passing to overcome key limitations, such as the reliance on labeled data and the inability to capture deeper anomaly patterns. Experimental results show that our method can effectively generalize to unseen samples to improve performance. In the future, we plan to extend the GFM to a broader range of fields, exploring various potential applications.

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A Related Work

A.1 Unsupervised Graph Anomaly Detection

Unsupervised Graph Anomaly Detection (UGAD) has recently advanced from classical heuristics toward sophisticated deep learning methods [47] that better harness structure and semantic irregularities in graph data [54]. A comprehensive survey highlights the evolution of deep UGAD techniques and underscores emerging research trends [42]. For single-task anomaly detection, CoLA [35] samples contrastive instance pairs to capture node-neighbor relationships and evaluates abnormality through statistical estimation. Similarly, TAM [41] further proposes learns tailored node representations for anomaly measure by maximizing the local affinity of nodes to their neighbors. Subsequently, HCM-A [18] combines self-supervised learning to predict the shortest path lengths between

pairs of nodes with Bayesian learning to train HCM, capturing the uncertainty of the parameters and preventing overfitting. GADAM [6] further addresses the conflict between local inconsistency mining. CARE [62] enhances the adjacency matrix with pseudo-labels to capture both local and global node affinities without strong graph assumptions to address potential biases from pseudo-labels. Among the most recent innovations, GAAN [23] tackles the problem of hyperparameter leakage in self-supervised GAD. Another notable direction explores the integration of hyperbolic geometry with high-order structure modeling. HC-GLAD [11] introduces a dual hyperbolic contrastive learning framework, leveraging hypergraphs constructed from motif-based groupings and embedding them in hyperbolic space to capture hierarchy and power-law characteristics. Moreover, SDGG [1] presents a self-discriminative modeling using only normal graphs and pseudo-anomalous graphs generated by a jointly trained model, which avoids the need for true anomalous graphs and provides new insights for anomaly detection. For unified detection methods, BOURNE [30] unifies node- and edge-level anomaly detection by constructing dual hypergraph views and measuring embedding agreement, eliminating the need for negative sampling and improving efficiency on large graphs. Similarly, UNIGAD [28] Maximum Rayleigh Quotient Subgraph Sampler that unifies multi-level formats by transferring objects at each level into graph-level tasks on subgraphs. Moreover, Uniform [45], leveraging an energy-based GNN for iterative multi-granular anomaly detection, unifies multiple tasks from a subgraph perspective.

However, these advanced methods are inherently dataset dependent, requiring separate training for each dataset and thus cannot generalize across multiple datasets from a single training. In contrast, our method can generalize across different datasets with only a single training, significantly reducing the training requirements while maintaining strong performance under varying conditions.

A.2 Graph Foundation Models

Recent efforts in graph machine learning have increasingly turned to the concept of GFMs, pre-trained models on extensive and diverse graph data that can generalize across a spectrum of downstream tasks. The work of [?] introduced GFMs and provided a rigorous taxonomy, categorizing existing approaches into GNN-based [13, 17], LLM-based [7, 14, 56], and GNN+LLM hybrid models. Building on this foundation, [49] presented a comprehensive survey that frames GFMs within a modular paradigm comprising backbone architectures, pretraining strategies, and adaptation mechanisms. Beyond surveys, Opengraph [51] leverages LLM-based data augmentation to mitigate data scarcity, introducing a unified graph tokenizer for robust generalization. Moreover, recent empirical benchmarks and models offer actionable insights. LangGFM [27] shows that pure LLM-based GFMs leveraging graph textualization and language-space augmentations can outperform specialized methods across a 26-dataset benchmark [55]. Similarly, the Text-space GFMs [8] provide a standardized benchmark to assess practical performance across domains. For GCN-based graph models, early approaches such as PT-HGNN [19] advance pre-training on large-scale heterogeneous graphs by jointly designing node- and schema-level contrastive tasks to capture transferable semantics. Similarly, AdapterGNN [22] introduces highly expressive adapters

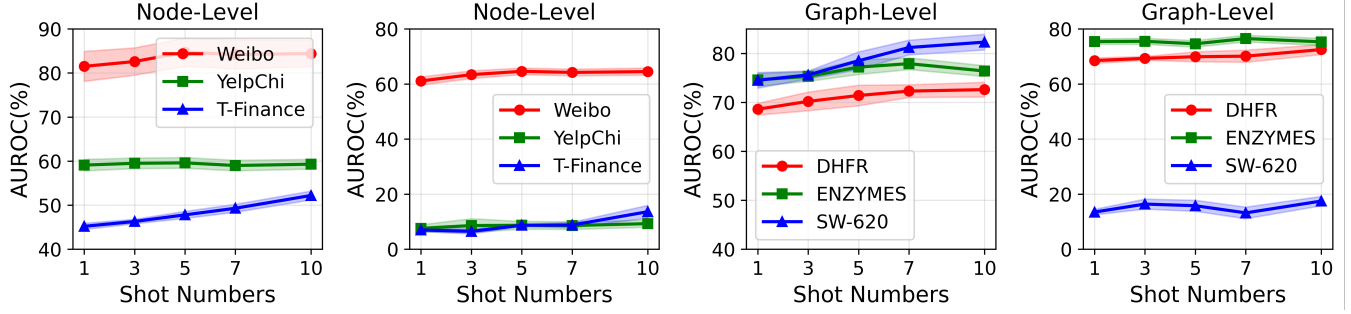


Figure 6: AUROC and AUPRC Performance of GFM-UAD under different numbers of shots for node-level tasks on Weibo, YelpChi, T-Finance, and graph-level tasks on DHFR, ENZYMES, and SW-620 datasets.

for GNNs that preserve the knowledge of large pretrained models, enabling effective adaptation to downstream tasks with only a small number of parameters. AnyGraph [50] proposes a unified graph model to address critical challenges, including structural heterogeneity, feature heterogeneity, and the need for rapid adaptation. GrapAny [61] inference on unseen graphs as an analytical solution to a LinearGNN and further employs an attention mechanism over multiple LinearGNNs to enhance generalization. Moreover, AffinityTune [5] pre-trains graph encoders to align learnable, data-independent normal and abnormal class prototypes with node representation residuals. UNPrompt [39], which leverages latent node attribute predictability as a generalized anomaly measure and learns universal graph patterns through a normalized node attribute space. ARC [30] directly extracts dataset-specific patterns from the target dataset using few-shot normal samples during inference. More recently, AnomalyGFM [40] pre-trains graph encoders to align data-independent normal and abnormal class prototypes with node representation residuals.

However, most of these methods fail to generalize effectively to multi-task anomaly detection. Even those that can generalize to single anomaly detection still rely on labeled data. In practice, obtaining labeled data is often difficult, which limits its applicability in real-world scenarios.

B Computation of Evaluation Metrics

In this work, we employ two standard metrics for graph anomaly detection: the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) and the Area Under the Precision-Recall Curve (AUC-PR). AUC-ROC measures the trade-off between true positive rate and false positive rate, while AUC-PR emphasizes the balance between precision and recall, which is particularly suitable for imbalanced anomaly detection scenarios. Together, these two metrics provide a comprehensive assessment of the detection performance. Their computation details are as follows:

AUROC Given the set of true labels $\{y_i\}$ and predicted anomaly scores $\{\hat{y}_i\}$ for all graphs, we compute the true positive rate (TPR) and false positive rate (FPR) at various thresholds τ : $TPR(\tau) = \frac{TP(\tau)}{TP(\tau) + FN(\tau)}$, $FPR(\tau) = \frac{FP(\tau)}{FP(\tau) + TN(\tau)}$, where TP, FP, TN, and FN denote the counts of true positives, false positives, true negatives, and false negatives at threshold τ . The ROC curve plots TPR against FPR, and the AUC-ROC is calculated as the area under this curve.

AUPRC Precision and recall are computed as: $Pre(\tau) = \frac{TP(\tau)}{TP(\tau) + FP(\tau)}$. The PR curve plots precision against recall across different thresholds, and the AUC-PR is obtained as the area under this curve. AUC-PR is particularly informative in highly imbalanced scenarios where anomalous graphs are rare. Both metrics provide threshold-independent evaluation of detection performance and are widely used in graph anomaly detection benchmarks.

C Details of Baseline Methods

- AnomalyDAE [10] addresses the complex cross-modality interactions between network structure and node attributes, which are often neglected by existing methods.

Table 5: AUROC results for edge-level tasks on five real-world single graph datasets with the models trained on Facebook only. For each dataset, the best performance within each metric is boldfaced, with the second-best underlined.

Methods	Amazon	Reddit	Weibo	YelpChi	T-Finance
BOURNE	66.7 $\pm$ 3.0	61.5 $\pm$ 2.8	70.4 $\pm$ 3.1	59.6 $\pm$ 3.2	OOM
RGAE	65.4 $\pm$ 1.9	66.6 $\pm$ 2.2	73.6 $\pm$ 1.6	61.2 $\pm$ 1.5	OOM
UniGAD	52.4 $\pm$ 1.6	57.8 $\pm$ 1.5	72.5 $\pm$ 1.5	66.8 $\pm$ 1.4	OOM
Uniform	50.7 $\pm$ 1.6	56.4 $\pm$ 0.6	70.3 $\pm$ 1.6	65.9 $\pm$ 1.5	55.8 $\pm$ 3.2
UNPrompt	56.3 $\pm$ 1.4	62.6 $\pm$ 1.8	68.6 $\pm$ 1.2	60.9 $\pm$ 1.3	OOM
AnomalyGFM	57.5 $\pm$ 1.6	63.4 $\pm$ 1.3	65.7 $\pm$ 2.3	63.4 $\pm$ 1.2	50.7 $\pm$ 1.5
OURS	71.2$\pm$1.4	69.8$\pm$1.6	76.8$\pm$1.5	71.4$\pm$1.3	62.4$\pm$1.3

D Details of Benchmark Datasets

In this paper, we use both single and multiple graph benchmark datasets. The parameters for these datasets are presented in Table 1. The detailed information for these datasets is as follows:

- **Amazon** [28]: This dataset is derived from Amazon product co-purchasing networks. Nodes represent product descriptions, and edges indicate frequent co-purchases.
- **Reddit** [28]: This dataset is constructed from Reddit posts and comments, where nodes represent users or threads, and edges indicate replies or interactions. Node features include text embeddings from posts.

Table 6: AUROC results for node-level task on six multiple graph datasets with the models trained on AIDS only. For each dataset, the best performance within each metric is boldfaced, with the second-best underlined.

Methods	MUTAG	BZR	COX2	DHFR	ENZYMES	SW-620
GADAM	61.7 $\pm$ 0.9	65.5 $\pm$ 1.2	64.8 $\pm$ 1.3	71.4 $\pm$ 1.2	63.1 $\pm$ 1.7	OOM
CARE	76.2 $\pm$ 1.2	67.9 $\pm$ 1.3	66.9 $\pm$ 1.3	71.7 $\pm$ 1.3	62.5 $\pm$ 1.6	OOM
UniGAD	59.6 $\pm$ 1.2	53.7 $\pm$ 0.8	49.3 $\pm$ 1.2	58.9 $\pm$ 2.4	64.4 $\pm$ 1.7	50.5 $\pm$ 1.4
Uniform	60.3 $\pm$ 1.0	66.4 $\pm$ 1.5	59.5 $\pm$ 2.1	62.3 $\pm$ 1.6	52.4 $\pm$ 1.8	OOM
UNPrompt	50.1 $\pm$ 1.4	49.4 $\pm$ 1.2	63.5 $\pm$ 2.0	63.6 $\pm$ 1.4	57.1 $\pm$ 1.2	52.6 $\pm$ 1.9
AnomalyGFM	72.4 $\pm$ 2.2	61.7 $\pm$ 0.3	70.6 $\pm$ 1.5	50.3 $\pm$ 0.6	58.9 $\pm$ 1.5	54.7 $\pm$ 1.6
ARC	75.4 $\pm$ 1.5	58.5 $\pm$ 1.4	66.7 $\pm$ 1.8	67.2 $\pm$ 1.5	58.9 $\pm$ 1.5	OOM
OURS	81.2$\pm$1.3	68.2$\pm$2.0	71.5$\pm$1.9	73.1$\pm$2.0	67.8$\pm$1.9	58.2$\pm$1.0

- **Weibo** [28]: This dataset is constructed from the social media platform Weibo, where nodes represent users. Edges denote following and follower relationships, and node attributes include user posts, demographics, or activity logs.
- **YelpChi** [28]: This dataset consists of a collection of Yelp reviews with associated metadata, capturing the characteristics of businesses across different cities.
- **Amazon-all** [28]: This dataset provides a collection of product reviews from the Amazon e-commerce platform.
- **YelpChi-all** [28]: This dataset extends the YelpChi dataset by incorporating a broader and more diverse scope of businesses across various categories.
- **Disney** [38]: This dataset is typically a user interaction dataset related to Disney platforms. Nodes may represent movies or TV shows, and edges capture co-watch behavior or genre similarities. Node features include metadata.
- **T-Finance** [40]: This dataset contains financial transactions. The nodes typically represent individual financial entities.
- **T-Social** [40]: This dataset contains data on social interactions. The nodes represent users, while the edges represent relationships.
- **MUTAG** [2]: This dataset consists of 188 graphs that represent chemical compounds. In this dataset, each graph corresponds to a molecular structure, where nodes represent atoms and edges represent chemical bonds.
- **BZR** [2]: This dataset contains 405 molecular graphs representing benzodiazepine receptor ligands. Nodes correspond to atoms with associated chemical features.
- **COX2** [4]: This dataset includes 467 molecular graphs of COX-2 inhibitors. Nodes represent atoms with 3-dimensional features, and edges represent chemical bonds.
- **DHFR** [29]: This dataset consists of molecular graphs of dihydrofolate reductase inhibitors. Each graph encodes a molecule, with nodes representing atoms and edges representing chemical bonds. Node features include atom types.
- **ENZYMES** [29]: This dataset consists of 600 protein tertiary structure graphs, each representing an enzyme classified into one of six classes. Nodes denote amino acid residues, and edges indicate spatial proximity.

- **PROTEINS** [4]: This dataset contains protein graphs where nodes represent secondary structure elements, and edges indicate neighborhood relationships.
- **DD** [1]: This dataset comprises 1,178 graphs representing protein structures. Nodes are amino acids, and edges denote spatial proximity.
- **SW-620** [53]: This dataset consists of 40,532 graphs, each representing a sample from a cancerous cell line. The graphs represent cell signaling pathways, with nodes corresponding to molecules, and edges indicating interactions between molecules.
- **PC-3** [53]: This dataset represents molecular structures related to a prostate cancer cell line. Each graph corresponds to a chemical compound with nodes representing atoms.

E Implementation Details

To ensure the completeness and reproducibility of the experiments, we provide details in the following four aspects: hardware resources, software resources, parameter settings, and baseline adaptation.

Hardware Resources We train the models on a single NVIDIA GeForce RTX 4090 GPU, powered by an Intel Core i9-12900K 12th-generation CPU and equipped with 64GB of Kingston DDR4 RAM.

Software Resources All experiments are implemented in Python using PyTorch (version 2.4.0, CUDA 11.7) and PyTorch Geometric (version 2.1.0), running on the Windows 10 Operating System.

Parameter Settings For optimization, we employ the Adam optimizer with a learning rate of 0.001, decaying at a rate of $\times 10^{-4}$, and utilize a batch size of 256. θ is fixed at 10. ζ is fixed to 4. σ is the ReLU activation function. Following common practice, we set the number of layers of the encoder to 4, with the hidden feature dimension fixed at 10 for the single graph dataset and 4 for multiple graph datasets. The model is trained over 100 epochs, ensuring sufficient convergence for each task. To ensure the robustness and reliability of the results, each experiment is independently repeated five times, with results averaged to mitigate potential variability.

Baseline Adaptation For cross-domain models designed for single-task applications (i.e., ARC, UNprompt, and AnomalyGFM), we align and adapt them to accommodate tasks at various levels of detection. For supervised anomaly detection models, we transform them into unsupervised approaches. Notably, the use of BOURNE as an edge-level task detection method follows the setup established in previous work by Uniform [45].

F More Experimental Results

F.1 Comparison Experiments

The experimental results for edge-level tasks on single-graph datasets and node-level tasks on multiple graph datasets are presented in Tables 5 and 6, respectively, which further demonstrate the superior performance of our method.

F.2 Hyper-parameters T Sensitivity Analysis

Experimental performance of GFM-UAD under different numbers of shots on Weibo, YelpChi, T-Finance, DHFR, ENZYMES, and SW-620 datasets, as shown in Fig. 6.